range of noble gas, moreover for this element both IE_1 and IE_2 are higher and hence X is the noble gas.

For Y, the first ionisation energy is low and second ionisation energy is very high and hence Y is most reactive metal.

For Z, both IE_1 and IE_2 are higher and hence it is least reactive.

7. $Cl(g) + e^- \rightarrow Cl^-(g)$ $\Delta H = 348$ kJ mol^{-1}

For one mole (35.5g) 348 kJ is released.

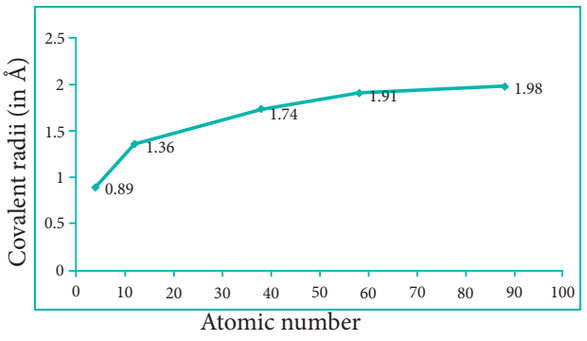
$\therefore$ For 17.75g chlorine, $\frac{348 \text{ kJ}}{35.5 \cancel{g}} \times 17.75 \cancel{g}$ energy leased.

$\therefore$ The amount of energy released $= \frac{348}{2}$

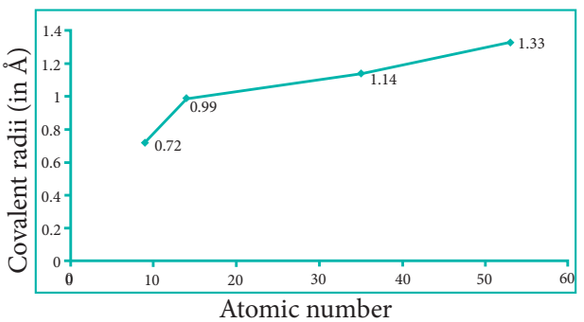
$= 174$ kJ

Activitiy : 3.1

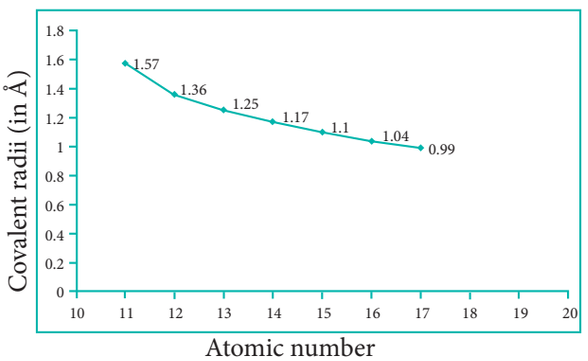
Covalent radii - 2nd group elements



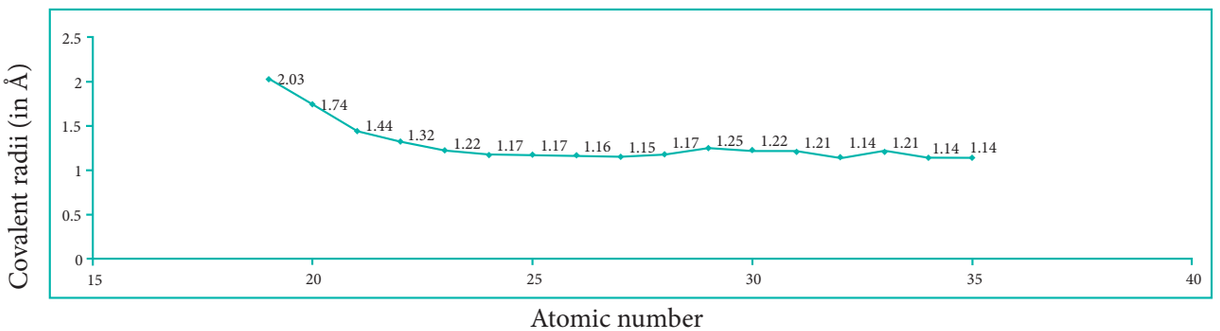
Covalent radii - 17th group elements



Covalent radii - 3rd period elements



Covalent radii - 4th period elements



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